

SBMS Module-Wise Implementation Playbook

Handbook reorganized by module so every module has its tables, forms, APIs, UI pages, workflows, validations and implementation checklist in one place.

Source: handbook.md uploaded by user | Generated for SBMS manual implementation

How to Use This Playbook

1. Pick the module you are implementing.
2. Open that module section only.
3. Complete work in this order: tables -> entity -> repository/HQL -> service validation -> controller -> Angular model/service/pages -> dashboard -> permissions -> testing.
4. Do not move to next module until the current module CRUD/workflow/dashboard/permission/testing are done.

Global Rules Reused by Every Module

Category	Details
Backend flow	Controller -> Service -> Repository -> Database. Concrete controller lives in controller.impl and service contains business rules.
Backend package pattern	com.sbms.<module>/{controller, controller/impl, service, service/impl, repository, entity, dto, enums, util}
Persistence style	Repository class + EntityManager + HQL only. Avoid JpaRepository as primary business repository style.
DI style	@Autowired preferred.
Validation style	Three layers: frontend form validation, frontend component.ts validation, backend service validation.
Response/exception	Use ApiResponse, ResponseBuilder, BusinessException/BadRequestException, ResourceNotFoundException, GlobalExceptionHandler.
Frontend pattern	src/app/features/<module>/{pages, models, services, <module>-routing.module.ts, <module>.module.ts}
List screen rule	Every major list must have Action column. View is mandatory; add Edit, Archive/Delete, Restore, Approve/Reject/Return, Print/Export, History as needed.
Dashboard rule	Every module needs dashboard cards, recent activity, status distribution and action queue.
Soft delete mindset	Master/config data usually soft delete/archive; transaction/ledger/audit/history data kept forever; some workflow records use status transitions only.

Enterprise Build Order by Sprint

Sprint	Scope
Phase 0	Project setup, DB foundation, common packages, auth groundwork, RBAC, Angular shell/layout, SweetAlert2,

	shared DataTable, Font Awesome, guards/interceptors.
Sprint 1	Role, permission, user, login/session, lookup/config, dashboard skeleton.
Sprint 2	Branch management, branch assignment, teller limit, vault balance, cash ledger, branch dashboard.
Sprint 3	ATM/CDM terminal, cash cassette/bin, replenishment, reconciliation, terminal dashboard.
Sprint 4	Customer, address, identity, KYC preparation, document tracking, customer dashboard.
Sprint 5	KYC flow, account types, account opening request, account approval, accounts, account dashboard.
Sprint 6	Deposit, withdrawal, fund transfer, transaction journal, reversal, transaction dashboard.
Sprint 7	Profit ratio, profit applicability, schedule, posting, balance snapshot, profit dashboard.
Sprint 8	Card, lifecycle, PIN event log, ATM/CDM card linkage, card dashboard.
Sprint 9	Customer statement, branch statement, export/print, statement dashboard.
Sprint 10	Deposit schemes, enrollment, schedule, statement, scheme dashboard.
Sprint 11	Financing products/applications, asset verification, approval chain, disbursement, repayment schedule.
Sprint 12	Contracts, template, clauses, versioning, signature states, contract dashboard.
Sprint 13	Shariah review, review cases, checklist, decision workflow, Shariah dashboard.
Sprint 14	Zakat profile, calculation, charity fund, beneficiary payout, zakat dashboard.
Sprint 15	Notifications, templates, delivery logs, integrations, retry log, integration dashboard.
Sprint 16	Reporting, regulatory reports, operational reports, print/export packs, report dashboard.
Sprint 17	Security event log, audit log, compliance case, suspicious activity, investigation dashboard.
Sprint 18	Workflow timeline, comments, attachments, dashboards polish, RBAC polish, E2E testing.

Cross-Module Dependency Map

Category	Details
Level 1 foundational	Auth, role, permission, lookup/config, branch, notification/integration base
Level 2 customer onboarding	Customer, customer_address, customer_document, KYC, verification
Level 3 operational account	Account type, account opening request, account, profit ratio, transaction, statements
Level 4 expanded operations	ATM/CDM, card, deposit scheme
Level 5 advanced approval	Financing, contract, Shariah review, calculation engine
Level 6 governance/support	Zakat/charity, reporting, security/compliance, dashboards/analytics polish

Module Index

#	Module	Sprint/Phase	Base API Path
1	Module 1: Role & Permission Management	Sprint 1	/api/roles
2	Module 2: User Management	Sprint 1	/api/users
3	Module 3: Lookup / Config	Sprint 1	/api/lookups
4	Module 4: Branch Management	Sprint 2	/api/branches
5	Module 5: ATM / CDM Management	Sprint 3	/api/atm-terminals
6	Module 6: Customer Management	Sprint 4	/api/customers
7	Module 7: KYC Management	Sprint 5	/api/kyc
8	Module 8: Islamic Account Management	Sprint 5	/api/account-types, /api/account-opening-requests, /api/accounts
9	Module 9: Deposit / Withdrawal / Transfer / Transactions	Sprint 6	/api/transactions
10	Module 10: Profit Ratio & Posting	Sprint 7	/api/profit-ratios, /api/profit-schedules, /api/profit-postings
11	Module 11: Card Management	Sprint 8	/api/cards
12	Module 12: Customer Statement / Branch Statement	Sprint 9	/api/customer-statements, /api/branch-statements
13	Module 13: Deposit Scheme	Sprint 10	/api/deposit-schemes
14	Module 14: Financing	Sprint 11	/api/financing-products, /api/financing-applications
15	Module 15: Contract Management	Sprint 12	/api/contracts
16	Module 16: Shariah Review	Sprint 13	/api/shariah-reviews
17	Module 17: Zakat & Charity	Sprint 14	/api/zakat
18	Module 18: Notification & Alerts	Sprint 15	/api/notifications
19	Module 19: Integration Management	Sprint 15	/api/integrations
20	Module 20: Reporting & Regulatory	Sprint 16	/api/reports
21	Module 21: Security / Audit / Investigation	Sprint 17	/api/security-events, /api/audit-logs, /api/investigations
22	Module 22: Workflow Support	Sprint 18	/api/workflows
23	Add-on Module: Verification / OTP / Email / SMS	Phase 0 + Sprint 1-2 + Sprint 4 + Sprint 15 + Sprint 17	/api/verification plus auth/customer/notification/integration hooks
24	Support Concept: Calculation Engine	Advanced support for Financing/Profit/Deposit Scheme	Internal service / optional /api/calculations

Reusable Implementation Checklist for Every Module

5. Create all master/child/workflow tables with PK/FK/unique/audit columns.
6. Create entity classes aligned with table/column names.
7. Create repository HQL methods: save, update, getById, list/search/filter, status-based search, soft delete/status change if allowed.
8. Create service validations: uniqueness, existence, workflow, branch/role/permission and business rules.
9. Create controller endpoints: create, update, getById, list/search, status/workflow action, dashboard summary.
10. Create Angular model/service, dashboard page, list page, create/edit/view pages and workflow pages.
11. Add DataTable-style search/filter/pagination/export, SweetAlert confirmations, status badges and action buttons.
12. Map permissions and hide/guard routes/buttons while enforcing backend checks too.
13. Test valid cases, invalid cases, role-wise access and end-to-end workflow.

Module 1: Role & Permission Management

Category	Details
Sprint/Phase	Sprint 1
Base API path	/api/roles
Goal	Manage roles and permission mapping. Role controls who can access module/action-level features.

Dashboard Widgets / Summary Cards

- Total roles
- Active roles
- Inactive/deleted roles
- Permission-heavy roles
- Recent role/permission changes

Tables & Columns

role

- id
- role_code
- role_name
- description
- status
- created_at
- updated_at
- created_by
- updated_by

role_permission

- id
- role_id
- permission_id
- allow_flag
- created_at
- created_by

Forms / Field Source / Save Mapping

Role create/edit form

Field	UI Type	Source	Save Target
Role Code	Text box or auto-generated	manual/system	role.role_code
Role Name	Text box	manual	role.role_name
Description	Textarea	manual	role.description
Status	Dropdown	ACTIVE/INACTIVE	role.status

Role permission mapping form

Field	UI Type	Source	Save Target
Role	Dropdown	role table	role_permission.role_id
Module	Dropdown	app_module table	permissions filtered by module
Actions	Checkbox list	app_permission	role_permission.permission_id

Backend Files / Package Guide

- com.sbms.role.controller/IRoleController.java
- com.sbms.role.controller.impl/RoleController.java
- com.sbms.role.service/IRoleService.java
- com.sbms.role.service.impl/RoleService.java
- com.sbms.role.repository/RoleRepository.java
- com.sbms.role.repository/RolePermissionRepository.java
- com.sbms.role.entity/Role.java
- com.sbms.role.entity/RolePermission.java
- com.sbms.role.dto/RolePermissionAssignDto.java

Angular Pages & Files

Category	Details
Pages	• role-dashboard • role-list • role-create • role-edit • role-view • role-permission-map
Core Angular files	• models/role.model.ts • models/role-permission.model.ts • services/role.service.ts • role-routing.module.ts • role.module.ts

Suggested APIs

- POST /api/roles/create
- GET /api/roles/list
- GET /api/roles/{id}
- PUT /api/roles/{id}
- DELETE /api/roles/{id}
- PUT /api/roles/{id}/restore
- GET /api/roles/dropdown
- GET /api/roles/permissions/{roleId}
- POST /api/roles/{id}/permissions/map

Business Flow / Workflow

14. System admin creates role
15. System admin maps permissions to role
16. Role is assigned to users
17. Menu/action visibility follows permissions

Important Validations / Business Rules

- role_code must be unique
- role_name should not duplicate logically
- Cannot delete/archive a role if assigned to active users
- SYSTEM_ADMIN role should be protected

View Page Requirement

Role basic info, mapped permission matrix, module/action list, status, audit info.

Action Column Buttons

- View
- Edit
- Archive/Delete
- Restore
- Map permissions

Implementation Order for This Module

18. Create role
19. List roles
20. Edit role
21. Map permissions
22. View role detail
23. Activate/Inactivate role

Module 2: User Management

Category	Details
Sprint/Phase	Sprint 1
Base API path	/api/users
Goal	Manage system users, branch mapping, login readiness, role assignment, lock/unlock and status control.

Dashboard Widgets / Summary Cards

- Total users
- Active users
- Locked users
- Users by role
- Users by branch
- Last login list

Tables & Columns

app_user

- id
- user_code
- username
- full_name
- email
- mobile
- password_hash
- branch_id
- employee_no
- designation
- is_locked
- is_active
- last_login_at
- failed_login_count
- status
- created_at
- updated_at
- created_by
- updated_by

user_role

- id
- user_id
- role_id
- created_at
- created_by

user_session

- id
- user_id
- login_time
- logout_time

- jwt_id
- ip_address
- device_info
- status

Forms / Field Source / Save Mapping

User create/edit form

Field	UI Type	Source	Save Target
User Code	Auto-generated	system	app_user.user_code
Username	Text box	manual	app_user.username
Full Name	Text box	manual	app_user.full_name
Email	Text box	manual	app_user.email
Mobile	Text box	manual	app_user.mobile
Employee No	Text box	manual	app_user.employee_no
Designation	Text box	manual	app_user.designation
Branch	Dropdown/by-search	branch table	app_user.branch_id
Roles	Checkbox/multi-select	role table	user_role.role_id
Password	Password field	manual	app_user.password_hash
Confirm Password	Password field	manual temp	not saved
Active	Checkbox	manual	app_user.is_active
Locked	Checkbox	manual	app_user.is_locked
Status	Dropdown	ACTIVE/INACTIVE	app_user.status

Backend Files / Package Guide

- com.sbms.user.controller/IUserController.java
- com.sbms.user.controller.impl/UserController.java
- com.sbms.user.service/IUserService.java
- com.sbms.user.service.impl/UserService.java
- com.sbms.user.repository/UserRepository.java
- com.sbms.user.entity/AppUser.java
- com.sbms.user.entity/UserRole.java
- com.sbms.user.entity/UserSession.java
- com.sbms.user.dto/UserRequestDto.java
- com.sbms.user.dto/UserResponseDto.java

Angular Pages & Files

Category	Details
Pages	• user-dashboard • user-list • user-create • user-edit • user-view • user-role-assign • user-password-reset • user-lock-unlock
Core Angular files	• models/user.model.ts • services/user.service.ts • user-routing.module.ts • user.module.ts

Suggested APIs

- POST /api/users/create
- GET /api/users/list
- GET /api/users/{id}
- PUT /api/users/{id}
- DELETE /api/users/{id}
- PUT /api/users/{id}/restore
- GET /api/users/dropdown
- GET /api/users/search
- POST /api/users/{id}/lock
- POST /api/users/{id}/unlock
- POST /api/users/{id}/reset-password
- POST /api/users/{id}/assign-role
- GET /api/users/{id}/history

Business Flow / Workflow

24. Admin creates user
25. Assign branch if branch-bound
26. Assign one or more roles
27. User logs in and receives role/permission set
28. Security/session events stored

Important Validations / Business Rules

- username unique
- email unique
- branch required for branch staff roles
- customer role cannot get admin permissions
- password hash only saved, never plain text
- inactive or locked user cannot log in

View Page Requirement

User profile, mapped roles, assigned branch, contact verification status, account status, login/security history summary.

Action Column Buttons

- View
- Edit
- Archive/Delete
- Restore
- Lock
- Unlock
- Reset password
- Assign role

Implementation Order for This Module

29. Create user
30. Assign role
31. List user
32. Edit user
33. Lock/unlock user

34. Activate/inactivate user
35. Reset password
36. User detail view

Module 3: Lookup / Config

Category	Details
Sprint/Phase	Sprint 1
Base API path	/api/lookups
Goal	Manage reusable dropdown/configuration data used by other modules.

Dashboard Widgets / Summary Cards

- Lookup type count
- Lookup value count
- Active values
- Recently changed values

Tables & Columns

lookup_type

- id
- type_code
- type_name
- description
- status
- created_at
- updated_at

lookup_value

- id
- lookup_type_id
- value_code
- value_label
- value_bn_label
- sort_order
- extra_data
- status
- created_at
- updated_at

Forms / Field Source / Save Mapping

Lookup type/value forms

Field	UI Type	Source	Save Target
Type Code	Text box	manual	lookup_type.type_code
Type Name	Text box	manual	lookup_type.type_name
Value Code	Text box	manual	lookup_value.value_code
Value Label	Text box	manual	lookup_value.value_label
Bangla Label	Text box	manual	lookup_value.value_bn_label
Sort Order	Number	manual	lookup_value.sort_order
Extra Data	Textarea/JSON	manual	lookup_value.extra_data
Status	Dropdown	ACTIVE/INACTIVE	status

Backend Files / Package Guide

- com.sbms.lookup.controller/ILookupController.java
- com.sbms.lookup.controller.impl/LookupController.java
- com.sbms.lookup.service/ILookupService.java
- com.sbms.lookup.service.impl/LookupService.java
- com.sbms.lookup.repository/LookupRepository.java
- com.sbms.lookup.entity/LookupType.java
- com.sbms.lookup.entity/LookupValue.java

Angular Pages & Files

Category	Details
Pages	• lookup-dashboard • lookup-type-list • lookup-type-create • lookup-type-view • lookup-value-list • lookup-value-create • lookup-value-edit • lookup-value-view
Core Angular files	• models/lookup.model.ts • services/lookup.service.ts • lookup-routing.module.ts • lookup.module.ts

Suggested APIs

- POST /api/lookups/types/create
- GET /api/lookups/types/list
- GET /api/lookups/types/{id}
- PUT /api/lookups/types/{id}
- DELETE /api/lookups/types/{id}
- POST /api/lookups/values/create
- GET /api/lookups/values/list
- GET /api/lookups/values/{id}
- PUT /api/lookups/values/{id}
- DELETE /api/lookups/values/{id}
- GET /api/lookups/values/by-type/{typeCode}
- GET /api/lookups/dropdown/{typeCode}

Business Flow / Workflow

37. Seed common lookup types first
38. Feature modules consume lookup dropdown values
39. Admin updates labels/status/sort order

Important Validations / Business Rules

- type_code unique
- value_code unique within type
- sort_order numeric
- inactive values should not appear in active dropdowns

View Page Requirement

Lookup type info, child values, display order, status flags.

Action Column Buttons

- View
- Edit
- Archive/Delete
- Restore

Implementation Order for This Module

40. Create lookup type
41. List lookup types
42. Create lookup value
43. List lookup values
44. Edit value
45. Soft delete/archive value

Module 4: Branch Management

Category	Details
Sprint/Phase	Sprint 2
Base API path	/api/branches
Goal	Manage branches, branch ownership, staff assignment, teller limits, vault balances, cash ledger and branch visibility.

Dashboard Widgets / Summary Cards

- Total branches
- Active branches
- Branch cash position
- Pending assignments
- Teller limit alerts
- Today vault opening/closing status

Tables & Columns

branch

- id
- branch_code
- branch_name
- branch_short_name
- branch_type
- routing_no
- swift_code
- email
- mobile
- phone
- address_line1
- address_line2
- country_id
- division_id
- district_id
- upazila_id
- postal_code
- manager_user_id
- opened_date
- status
- created_at
- updated_at
- created_by
- updated_by

branch_user_assignment

- id
- branch_id
- user_id
- assignment_role
- from_date

- to_date
- is_primary
- status
- created_at

teller_limit

- id
- branch_id
- user_id
- limit_date
- daily_deposit_limit
- daily_withdraw_limit
- single_txn_limit
- approved_by
- approved_at
- status
- created_at

vault_balance

- id
- branch_id
- balance_date
- opening_balance
- total_cash_in
- total_cash_out
- closing_balance
- is_closed
- closed_by
- closed_at
- remarks
- status
- created_at

branch_cash_ledger

- id
- branch_id
- ledger_date
- entry_type
- source_type
- reference_no
- debit_amount
- credit_amount
- balance_after
- remarks
- created_at
- created_by

Forms / Field Source / Save Mapping

Branch create/edit form

Field	UI Type	Source	Save Target
Branch Code	Auto-generated or text	system/manual	branch.branch_code
Branch Name	Text box	manual	branch.branch_name
Short Name	Text box	manual	branch.branch_short_name
Branch Type	Dropdown	lookup BRANCH_TYPE	branch.branch_type
Routing No	Text box	manual	branch.routing_no
Swift Code	Text box	manual	branch.swift_code
Email	Text box	manual	branch.email
Mobile	Text box	manual	branch.mobile
Phone	Text box	manual	branch.phone
Address Line 1	Text box	manual	branch.address_line1
Address Line 2	Text box	manual	branch.address_line2
Country	Dropdown	country table	branch.country_id
Division	Dropdown	division table	branch.division_id
District	Dropdown	district table	branch.district_id
Upazila	Dropdown	upazila table	branch.upazila_id
Postal Code	Text box	manual	branch.postal_code
Manager	By-search dropdown	app_user	branch.manager_user_id
Opened Date	Date picker	manual	branch.opened_date
Status	Dropdown	ACTIVE/INACTIVE	branch.status

Branch user assignment form

Field	UI Type	Source	Save Target
Branch	Dropdown/search	branch	branch_user_assignment.branch_id
User	By-search	app_user	branch_user_assignment.user_id
Assignment Role	Dropdown	lookup ASSIGNMENT_ROLE	branch_user_assignment.assignment_role
From Date	Date picker	manual	branch_user_assignment.from_date
To Date	Date picker	manual	branch_user_assignment.to_date
Primary	Checkbox	manual	branch_user_assignment.is_primary
Status	Dropdown	ACTIVE/INACTIVE	branch_user_assignment.status

Teller limit form

Field	UI Type	Source	Save Target
Branch	Dropdown/search	branch	teller_limit.branch_id
Teller User	By-search	app_user filtered by branch	teller_limit.user_id
Limit Date	Date picker	manual	teller_limit.limit_date
Daily Deposit Limit	Currency	manual	teller_limit.daily_deposit_limit
Daily Withdraw Limit	Currency	manual	teller_limit.daily_withdraw_limit
Single Txn Limit	Currency	manual	teller_limit.single_txn_limit
Status	Dropdown	ACTIVE/INACTIVE	teller_limit.status

Vault opening form

Field	UI Type	Source	Save Target
-------	---------	--------	-------------

Branch	Dropdown/search	branch	vault_balance.branch_id
Balance Date	Date picker	manual	vault_balance.balance_date
Opening Balance	Currency	manual	vault_balance.opening_balance
Remarks	Textarea	manual	vault_balance.remarks

Backend Files / Package Guide

- com.sbms.branch.controller/IBranchController.java
- com.sbms.branch.controller.impl/BranchController.java
- com.sbms.branch.service/IBranchService.java
- com.sbms.branch.service.impl/BranchService.java
- com.sbms.branch.repository/BranchRepository.java
- com.sbms.branch.repository/BranchUserAssignmentRepository.java
- com.sbms.branch.repository/BranchTellerLimitRepository.java
- com.sbms.branch.repository/VaultBalanceRepository.java
- com.sbms.branch.repository/BranchCashLedgerRepository.java
- com.sbms.branch.entity/Branch.java
- BranchUserAssignment.java
- BranchTellerLimit.java
- VaultBalance.java
- BranchCashLedger.java
- dto package request/response DTOs

Angular Pages & Files

Category	Details
Pages	• branch-dashboard • branch-list • branch-create • branch-edit • branch-view • branch-user-assignment • teller-limit-list • teller-limit-create • teller-limit-view • vault-balance-list • vault-open • vault-view • cash-ledger-list • inter-branch-transfer • eod-summary
Core Angular files	• models/branch.model.ts • models/branch-assignment.model.ts • models/teller-limit.model.ts • services/branch-api.service.ts • services/branch-assignment-api.service.ts • services/teller-limit-api.service.ts • branch-routing.module.ts • branch.module.ts

Suggested APIs

- POST /api/branches/create
- GET /api/branches/list
- GET /api/branches/{id}

- PUT /api/branches/{id}
- DELETE /api/branches/{id}
- PUT /api/branches/{id}/restore
- GET /api/branches/dropdown
- POST /api/branches/assign-user OR POST /api/branches/assignments/create
- GET /api/branches/assignments
- GET /api/branches/assignments/{id}
- POST /api/branches/teller-limit/create
- GET /api/branches/teller-limit/list
- GET /api/branches/teller-limit/{id}
- PUT /api/branches/teller-limit/{id}
- POST /api/branches/vault/open
- POST /api/branches/vault/close
- GET /api/branches/vault/list
- GET /api/branches/vault/{id}
- GET /api/branches/cash-ledger
- GET /api/branches/dashboard-summary

Business Flow / Workflow

46. Admin creates branch
47. Branch manager assigned
48. Teller/staff assigned to branch
49. Teller limit configured
50. Vault opening balance created
51. Branch performs daily transactions
52. Cash/vault position updated
53. End-of-day summary generated

Important Validations / Business Rules

- branch_code unique
- one branch manager per period
- one vault opening per branch per date
- teller must belong to same branch for limit
- teller transaction cannot exceed teller limit
- cannot close vault twice
- inter-branch transfer needs approval trail
- EOD should reflect transaction totals correctly

View Page Requirement

Branch profile, manager and assigned users, teller limits, latest vault balance, latest cash ledger summary, EOD summary.

Action Column Buttons

- View
- Edit
- Archive/Delete
- Restore
- Assign user
- Open vault
- Close vault

- View ledger
- Start EOD

Implementation Order for This Module

54. Branch create
55. Branch list/edit/view
56. Branch user assignment
57. Teller limit create/edit
58. Vault opening
59. Cash ledger auto entries
60. Vault close
61. Branch dashboard

Module 5: ATM / CDM Management

Category	Details
Sprint/Phase	Sprint 3
Base API path	/api/atm-terminals
Goal	Manage ATM/CDM terminals, cash bins/cassettes, replenishment, reconciliation and terminal dashboard.

Dashboard Widgets / Summary Cards

- Total terminals
- Active terminals
- Cash low alert
- Unreconciled terminals
- Terminal downtime list
- Today ATM/CDM volumes

Tables & Columns

terminal

- id
- terminal_code
- terminal_name
- terminal_type
- branch_id
- location_note
- ip_address
- serial_no
- vendor_name
- install_date
- status
- created_at
- updated_at

terminal_cash_bin

- id
- terminal_id
- bin_no
- denomination
- max_capacity
- current_count
- status
- created_at

terminal_replenishment

- id
- terminal_id
- replenishment_date
- bin_no
- denomination
- quantity_added
- amount_added

- performed_by
- remarks
- status
- created_at

terminal_reconciliation

- id
- terminal_id
- recon_date
- system_amount
- physical_amount
- variance_amount
- approved_by
- approved_at
- remarks
- status
- created_at

Forms / Field Source / Save Mapping

Forms summary

Field	UI Type	Source	Save Target
Terminal form	Text/dropdown	branch + terminal type	terminal
Cash bin setup form	Number/currency	terminal	terminal_cash_bin
Replenishment form	Date/number/currency	terminal + app_user	terminal_replenishment
Reconciliation form	Currency/remarks	terminal	terminal_reconciliation

Backend Files / Package Guide

- com.sbms.atm.controller/IAtmTerminalController.java
- com.sbms.atm.controller.impl/AtmTerminalController.java
- service/repository/entity/dto for terminal, cash bin, replenishment, reconciliation

Angular Pages & Files

Category	Details
Pages	• terminal-dashboard • terminal-list • terminal-create • terminal-edit • terminal-view • cash-bin-list • cash-bin-create • cash-bin-view • replenishment-list • replenishment-create • replenishment-view • reconciliation-list • reconciliation-create • reconciliation-view • device-journal-list
Core Angular files	• models/terminal.model.ts • services/atm-terminal.service.ts • atm-routing.module.ts • atm.module.ts

Suggested APIs

- POST /api/atm-terminals/create
- GET /api/atm-terminals/list
- GET /api/atm-terminals/{id}
- PUT /api/atm-terminals/{id}
- DELETE /api/atm-terminals/{id}
- PUT /api/atm-terminals/{id}/restore
- GET /api/atm-terminals/dropdown
- POST /api/atm-terminals/cash-bin/create
- GET /api/atm-terminals/cash-bin/list
- GET /api/atm-terminals/cash-bin/{id}
- POST /api/atm-terminals/replenishment/create
- GET /api/atm-terminals/replenishment/list
- GET /api/atm-terminals/replenishment/{id}
- POST /api/atm-terminals/reconciliation/create
- GET /api/atm-terminals/reconciliation/list
- GET /api/atm-terminals/reconciliation/{id}
- GET /api/atm-terminals/device-journal
- GET /api/atm-terminals/dashboard-summary

Business Flow / Workflow

62. Admin registers terminal
63. Terminal mapped to branch
64. Cash cassette/bin setup created
65. ATM/CDM transactions recorded
66. Reconciliation compares terminal journal vs core records
67. Variance/exception handled

Important Validations / Business Rules

- terminal_code unique
- terminal must have branch mapping
- only active terminal can transact
- cash bin capacity should not be exceeded
- reconciliation variance must be tracked

View Page Requirement

Terminal profile, branch mapping, cash bins, latest replenishment, reconciliation and device journal summary.

Action Column Buttons

- View
- Edit
- Archive/Delete
- Restore
- Add cash bin
- Replenish
- Reconcile

Implementation Order for This Module

68. Create terminal tables

- 69. Terminal CRUD
- 70. Cash bin setup
- 71. Replenishment flow
- 72. Reconciliation flow
- 73. ATM dashboard

Module 6: Customer Management

Category	Details
Sprint/Phase	Sprint 4
Base API path	/api/customers
Goal	Store customer identity, profile, address, contact, branch and onboarding status.

Dashboard Widgets / Summary Cards

- Total customers
- Active customers
- Pending KYC customers
- Customers by branch
- New customers this month
- Incomplete profiles
- Blocked customers

Tables & Columns

customer

- id
- customer_code
- customer_type
- full_name
- father_name
- mother_name
- spouse_name
- date_of_birth
- gender
- marital_status
- nationality
- mobile
- email
- occupation
- monthly_income
- source_of_funds
- branch_id
- customer_status
- status
- created_at
- updated_at
- created_by
- updated_by

customer_address

- id
- customer_id
- address_type
- address_line1
- address_line2
- country_id

- division_id
- district_id
- upazila_id
- postal_code
- is_primary
- status
- created_at

customer_identity

- id
- customer_id
- document_type
- document_no
- issue_date
- expiry_date
- issue_country
- verified_flag
- status
- created_at

Forms / Field Source / Save Mapping

Customer onboarding form

Field	UI Type	Source	Save Target
Customer Code	Auto-generated	system	customer.customer_code
Customer Type	Dropdown	lookup CUSTOMER_TYPE	customer.customer_type
Full Name	Text box	manual	customer.full_name
Father Name	Text box	manual	customer.father_name
Mother Name	Text box	manual	customer.mother_name
Spouse Name	Text box	manual	customer.spouse_name
DOB	Date picker	manual	customer.date_of_birth
Gender	Dropdown	lookup GENDER	customer.gender
Marital Status	Dropdown	lookup MARITAL_STATUS	customer.marital_status
Nationality	Dropdown	country table	customer.nationality
Mobile	Text box	manual	customer.mobile
Email	Text box	manual	customer.email
Occupation	Text box	manual	customer.occupation
Monthly Income	Currency	manual	customer.monthly_income
Source of Funds	Textarea	manual	customer.source_of_funds
Branch	Dropdown/search	branch	customer.branch_id
Status	Dropdown	ACTIVE/PENDING	customer.status

Customer address form

Field	UI Type	Source	Save Target
Customer	By-search	customer	customer_address.customer_id
Address Type	Dropdown	lookup ADDRESS_TYPE	customer_address.address_type
Address Line 1	Text box	manual	customer_address.address_line1
Address Line 2	Text box	manual	customer_address.address_line2
Country	Dropdown	country	customer_address.country_id

			d
Division	Dropdown	division	customer_address.division_id
District	Dropdown	district	customer_address.district_id
Upazila	Dropdown	upazila	customer_address.upazila_id
Postal Code	Text box	manual	customer_address.postal_code
Primary	Checkbox	manual	customer_address.is_primary

Customer identity form

Field	UI Type	Source	Save Target
Customer	By-search	customer	customer_identity.customer_id
Document Type	Dropdown	lookup DOCUMENT_TYPE	customer_identity.document_type
Document No	Text box	manual	customer_identity.document_no
Issue Date	Date picker	manual	customer_identity.issue_date
Expiry Date	Date picker	manual	customer_identity.expiry_date
Issue Country	Dropdown	country	customer_identity.issue_country

Backend Files / Package Guide

- com.sbms.customer.controller/ICustomerController.java
- controller.impl/CustomerController.java
- service/repository/entity/dto for customer/address/identity

Angular Pages & Files

Category	Details
Pages	• customer-dashboard • customer-list • customer-create • customer-edit • customer-view • customer-address-manage • customer-identity-manage • customer-search • customer-status-action
Core Angular files	• models/customer.model.ts • services/customer.service.ts • customer-routing.module.ts • customer.module.ts

Suggested APIs

- POST /api/customers/create
- GET /api/customers/list
- GET /api/customers/{id}
- PUT /api/customers/{id}
- DELETE /api/customers/{id}

- PUT /api/customers/{id}/restore
- GET /api/customers/dropdown
- GET /api/customers/search
- POST /api/customers/address/create
- PUT /api/customers/address/{id}
- GET /api/customers/address/{id}
- POST /api/customers/identity/create
- PUT /api/customers/identity/{id}
- GET /api/customers/identity/{id}
- POST /api/customers/{id}/activate
- POST /api/customers/{id}/block
- GET /api/customers/{id}/timeline
- GET /api/customers/dashboard-summary

Business Flow / Workflow

74. Officer opens customer create form
75. Identity/contact/address captured
76. Customer code generated
77. Documents uploaded
78. Customer profile saved
79. KYC process starts

Important Validations / Business Rules

- customer_code unique
- at least one identity record needed for KYC
- NID/passport no should not duplicate wrongly
- mobile format check
- email/mobile uniqueness as needed
- customer cannot be activated before required checks

View Page Requirement

Customer profile, present/permanent address, identity docs, contact verification, KYC summary, accounts summary, card summary, recent transactions.

Action Column Buttons

- View
- Edit
- Archive/Delete
- Restore
- Activate
- Block
- Manage address
- Manage identity

Implementation Order for This Module

80. Create customer tables
81. Customer CRUD
82. Address CRUD
83. Identity CRUD

- 84. Customer search/detail
- 85. Customer dashboard

Module 7: KYC Management

Category	Details
Sprint/Phase	Sprint 5
Base API path	/api/kyc
Goal	Confirm customer eligibility, source of funds, documents, address, PEP/sanction/AML flags and risk level.

Dashboard Widgets / Summary Cards

- Pending KYC
- Verified KYC
- Rejected KYC
- Resubmission queue
- Risk level summary
- High-risk customers

Tables & Columns

kyc_profile

- id
- customer_id
- risk_level
- source_of_funds_note
- pep_flag
- sanction_flag
- aml_flag
- review_status
- reviewed_by
- reviewed_at
- remarks
- status
- created_at
- updated_at

customer_document

- id
- customer_id
- document_type
- file_reference_id
- document_no
- issue_date
- expiry_date
- verified_flag
- status
- created_at

kyc_decision_history

- id
- kyc_profile_id
- decision
- decision_by

- decision_at
- remarks
- status

Forms / Field Source / Save Mapping

KYC review form

Field	UI Type	Source	Save Target
Customer	By-search	customer	kyc_profile.customer_id
Risk Level	Dropdown	lookup RISK_LEVEL	kyc_profile.risk_level
Source of Funds Note	Textarea	manual	kyc_profile.source_of_funds_note
PEP Flag	Checkbox	manual	kyc_profile.pep_flag
Sanction Flag	Checkbox	manual	kyc_profile.sanction_flag
AML Flag	Checkbox	manual	kyc_profile.aml_flag
Remarks	Textarea	manual	kyc_profile.remarks
Review Status	Dropdown	PENDING/VERIFIED/ REJECTED/RESUBMIT	kyc_profile.review_status

Backend Files / Package Guide

- com.sbms.kyc.controller/KycController.java
- controller.impl/KycController.java
- service/repository/entity/dto for profile, documents, decision history

Angular Pages & Files

Category	Details
Pages	• kyc-dashboard • kyc-list • kyc-create • kyc-edit • kyc-view • kyc-review • kyc-approval-queue • kyc-document-upload • kyc-history
Core Angular files	• models/kyc.model.ts • services/kyc.service.ts • kyc-routing.module.ts • kyc.module.ts

Suggested APIs

- POST /api/kyc/profile/create
- GET /api/kyc/profile/list
- GET /api/kyc/profile/{id}
- PUT /api/kyc/profile/{id}
- POST /api/kyc/profile/{id}/submit
- POST /api/kyc/profile/{id}/verify
- POST /api/kyc/profile/{id}/approve
- POST /api/kyc/profile/{id}/reject
- POST /api/kyc/profile/{id}/return
- POST /api/kyc/document/upload
- GET /api/kyc/decision-history/{kycId}

- GET /api/kyc/profile/{id}/documents
- GET /api/kyc/dashboard-summary

Business Flow / Workflow

86. Customer profile exists
87. KYC profile created
88. Documents uploaded
89. Source of funds declared
90. Risk score captured
91. Reviewer verifies
92. Approve/reject/correction decision stored
93. Onboarding readiness updated

Status flow: DRAFT -> SUBMITTED -> UNDER_REVIEW -> VERIFIED -> APPROVED / REJECTED / SENT_BACK

Important Validations / Business Rules

- KYC cannot approve without required docs
- reviewer identity must be stored
- correction remarks mandatory on send back
- reviewer cannot approve own case if business policy requires
- risk level required

View Page Requirement

Customer + KYC profile, risk flags, documents, decision history, remarks and reviewer timestamps.

Action Column Buttons

- View
- Edit
- Submit
- Verify
- Approve
- Reject
- Return
- Upload document
- History

Implementation Order for This Module

94. Create KYC tables
95. KYC review flow
96. Document upload
97. Decision history
98. KYC dashboard

Module 8: Islamic Account Management

Category	Details
Sprint/Phase	Sprint 5
Base API path	/api/account-types, /api/account-opening-requests, /api/accounts
Goal	Handle account type setup, account opening request, verification, account creation, activation and profit linkage.

Dashboard Widgets / Summary Cards

- Total accounts
- Accounts by type
- Pending opening requests
- Active accounts
- Blocked/frozen accounts
- Accounts awaiting verification

Tables & Columns

account_type

- id
- type_code
- type_name
- shariah_contract_type
- currency_code
- minimum_opening_balance
- profit_applicable
- withdrawal_allowed
- status
- created_at

account_opening_request

- id
- request_no
- customer_id
- account_type_id
- branch_id
- requested_date
- initial_deposit_amount
- request_status
- verified_by
- verified_at
- approved_by
- approved_at
- remarks
- status
- created_at

account

- id
- account_number

- customer_id
- account_type_id
- branch_id
- opening_request_id
- opened_date
- current_balance
- available_balance
- profit_ratio_id
- account_status
- closed_date
- remarks
- status
- created_at
- updated_at

Forms / Field Source / Save Mapping

Account opening form

Field	UI Type	Source	Save Target
Request No	Auto-generated	system	account_opening_request.request_no
Customer	By-search	customer	account_opening_request.customer_id
Account Type	Dropdown	account_type	account_opening_request.account_type_id
Branch	Dropdown/search	branch	account_opening_request.branch_id
Initial Deposit	Currency	manual	account_opening_request.initial_deposit_amount
Requested Date	Date picker	manual/system	account_opening_request.requested_date
Remarks	Textarea	manual	account_opening_request.remarks

Backend Files / Package Guide

- com.sbms.account.controller/IAccountTypeController.java
- IAccountOpeningRequestController.java
- IAccountController.java
- service/repository/entity/dto for account type, opening request, account

Angular Pages & Files

Category	Details
Pages	• account-dashboard • account-type-list • account-type-create • account-type-view • account-opening-request-list • account-opening-request-create • account-opening-request-edit • account-opening-request-view • account-opening-review • account-list • account-view

	• account-status-action
Core Angular files	• models/account.model.ts • services/account.service.ts • account-routing.module.ts • account.module.ts

Suggested APIs

- POST /api/account-types/create
- GET /api/account-types/list
- GET /api/account-types/{id}
- PUT /api/account-types/{id}
- DELETE /api/account-types/{id}
- PUT /api/account-types/{id}/restore
- GET /api/account-types/dropdown
- POST /api/account-opening-requests/create
- GET /api/account-opening-requests/list
- GET /api/account-opening-requests/{id}
- PUT /api/account-opening-requests/{id}
- POST /api/account-opening-requests/{id}/submit
- POST /api/account-opening-requests/{id}/verify
- POST /api/account-opening-requests/{id}/approve
- POST /api/account-opening-requests/{id}/reject
- POST /api/account-opening-requests/{id}/return
- GET /api/accounts/list
- GET /api/accounts/{id}
- POST /api/accounts/{id}/activate
- POST /api/accounts/{id}/block
- POST /api/accounts/{id}/freeze
- POST /api/accounts/{id}/close
- GET /api/accounts/search
- GET /api/accounts/dashboard-summary

Business Flow / Workflow

99. Account types exist
100. Customer chooses account type
101. Opening request submitted
102. Teller preliminary review
103. Operations verification
104. Shariah compliance check
105. Account number generated
106. Account activated
107. Profit schedule/applicability linked

Status flow: DRAFT -> SUBMITTED -> PRELIM_CHECK -> OPS_VERIFIED -> SHARIAH_CHECKED -> APPROVED -> ACTIVE

Important Validations / Business Rules

- Customer must have approved KYC before final activation
- Account number must be system generated
- One request should not accidentally create duplicate active account
- Account type controls minimum balance/profit applicability

- Initial deposit must satisfy account type rules

View Page Requirement

Request detail, customer/KYC summary, account type rules, workflow history, created account detail.

Action Column Buttons

- View
- Edit request
- Submit
- Verify
- Approve
- Reject
- Return
- Activate account
- Block
- Freeze
- Close

Implementation Order for This Module

108.Create account type CRUD

109.Create account opening request flow

110.Account approval + account creation flow

111.Account lifecycle

112.Account dashboard

Module 9: Deposit / Withdrawal / Transfer / Transactions

Category	Details
Sprint/Phase	Sprint 6
Base API path	/api/transactions
Goal	Handle deposit, withdrawal, fund transfer, cheque/standing instructions, transaction journal and reversal.

Dashboard Widgets / Summary Cards

- Today deposit total
- Today withdrawal total
- Today transfer total
- Pending reversals
- Teller limit usage
- Suspicious transaction count
- Top branches by transaction count

Tables & Columns

transaction_journal

- id
- transaction_ref
- transaction_date
- transaction_type
- channel_type
- branch_id
- terminal_id
- debit_account_id
- credit_account_id
- amount
- narration
- posted_by
- approved_by
- reversal_flag
- parent_transaction_id
- transaction_status
- status
- created_at

cash_transaction

- id
- transaction_id
- cash_type
- cash_direction
- teller_user_id
- branch_id
- amount
- remarks
- created_at

fund_transfer

- id
- transaction_id
- from_account_id
- to_account_id
- transfer_mode
- remarks
- created_at

transaction_reversal

- id
- original_transaction_id
- reversal_transaction_id
- requested_by
- requested_at
- approved_by
- approved_at
- reason
- status
- created_at

Forms / Field Source / Save Mapping

Deposit form

Field	UI Type	Source	Save Target
Transaction Ref	Auto-generated	system	transaction_journal.transaction_ref
Branch	Dropdown/search	branch	transaction_journal.branch_id
Teller	Derived/logged-in	system	cash_transaction.teller_user_id
Account	By-search	account	transaction_journal.credit_account_id
Amount	Currency	manual	transaction_journal.amount
Narration	Textarea	manual	transaction_journal.narration
Txn Date	Datetime	system/manual	transaction_journal.transaction_date

Withdrawal form

Field	UI Type	Source	Save Target
Transaction Ref	Auto-generated	system	transaction_journal.transaction_ref
Branch	Dropdown/search	branch	transaction_journal.branch_id
Account	By-search	account	transaction_journal.debit_account_id
Amount	Currency	manual	transaction_journal.amount
Narration	Textarea	manual	transaction_journal.narration

Fund transfer form

Field	UI Type	Source	Save Target
-------	---------	--------	-------------

Transaction Ref	Auto-generated	system	transaction_journal.transaction_ref
From Account	By-search	account	fund_transfer.from_account_id
To Account	By-search	account	fund_transfer.to_account_id
Transfer Mode	Dropdown	lookup TRANSFER_MODE	fund_transfer.transfer_mode
Amount	Currency	manual	transaction_journal.amount
Remarks	Textarea	manual	fund_transfer.remarks

Backend Files / Package Guide

- com.sbms.transaction.controller/ITransactionController.java
- controller.impl/TransactionController.java
- service/repository/entity/dto for journal, cash transaction, fund transfer, reversal

Angular Pages & Files

Category	Details
Pages	• transaction-dashboard • cash-deposit • cash-withdraw • fund-transfer • cheque-clearing • standing-instruction-list • standing-instruction-create • transaction-list • transaction-view • transaction-reversal
Core Angular files	• models/transaction.model.ts • services/transaction.service.ts • transaction-routing.module.ts • transaction.module.ts

Suggested APIs

- POST /api/transactions/deposit
- POST /api/transactions/withdraw
- POST /api/transactions/transfer
- POST /api/transactions/cheque-clearing
- POST /api/transactions/standing-instruction/create
- GET /api/transactions/standing-instruction/list
- GET /api/transactions/list
- GET /api/transactions/{id}
- GET /api/transactions/search
- POST /api/transactions/{id}/reverse
- GET /api/transactions/{id}/journal
- GET /api/transactions/dashboard-summary

Business Flow / Workflow

- 113.Teller receives request
- 114.Account validated
- 115.Daily/teller limit checked
- 116.Business/Shariah check applied if needed
- 117.Transaction posted

- 118. Balance updated
- 119. Journal created
- 120. Alert sent
- 121. Statement/report updated

Important Validations / Business Rules

- Account must be active
- Withdrawal cannot exceed balance
- Teller limit check needed
- Transfer cannot be same account
- Branch ownership must be respected
- Reversal only by authorized role
- Transaction must be atomic
- Unique transaction_ref

View Page Requirement

Transaction header, debit/credit account, cash/fund details, journal, reversal status, audit trail.

Action Column Buttons

- View
- Reverse
- Print voucher
- Export
- Search

Implementation Order for This Module

- 122. Create transaction tables
- 123. Deposit form and posting
- 124. Withdrawal form and posting
- 125. Transfer form and posting
- 126. Reversal flow
- 127. Transaction dashboard

Module 10: Profit Ratio & Posting

Category	Details
Sprint/Phase	Sprint 7
Base API path	/api/profit-ratios, /api/profit-schedules, /api/profit-postings
Goal	Configure PSR/effective dates and post profit periodically with traceable schedules and postings.

Dashboard Widgets / Summary Cards

- Active profit ratios
- Pending posting cycles
- Posted this month
- Failed posting logs
- Current PSR table
- Upcoming posting run

Tables & Columns

profit_ratio

- id
- ratio_code
- account_type_id
- effective_from
- effective_to
- ratio_percent
- status
- created_at

profit_schedule

- id
- account_id
- profit_frequency
- next_posting_date
- last_posting_date
- status
- created_at

profit_posting

- id
- posting_ref
- account_id
- schedule_id
- posting_date
- profit_amount
- period_from
- period_to
- posted_by
- status
- created_at

balance_snapshot

- id
- account_id
- snapshot_date
- closing_balance
- average_balance
- status
- created_at

Forms / Field Source / Save Mapping

Profit forms

Field	UI Type	Source	Save Target
Ratio Code	Auto/text	system/manual	profit_ratio.ratio_code
Account Type	Dropdown	account_type	profit_ratio.account_type_id
Effective From/To	Date picker	manual	profit_ratio.effective_from/ effective_to
Ratio Percent	Decimal	manual	profit_ratio.ratio_percent
Profit Frequency	Dropdown	lookup PROFIT_FREQUENCY	profit_schedule.profit_frequency
Posting Run	Action	system	profit_posting

Backend Files / Package Guide

- com.sbms.profit.controller/IProfitRatioController.java
- IProfitScheduleController.java
- IProfitPostingController.java
- service/repository/entity/dto for ratio, schedule, posting, snapshot

Angular Pages & Files

Category	Details
Pages	• profit-dashboard • profit-ratio-list • profit-ratio-create • profit-ratio-view • profit-schedule-list • profit-schedule-create • profit-schedule-view • profit-posting-list • profit-posting-run • profit-posting-view
Core Angular files	• models/profit.model.ts • services/profit.service.ts • profit-routing.module.ts • profit.module.ts

Suggested APIs

- POST /api/profit-ratios/create
- GET /api/profit-ratios/list
- GET /api/profit-ratios/{id}
- PUT /api/profit-ratios/{id}
- DELETE /api/profit-ratios/{id}
- GET /api/profit-ratios/dropdown

- POST /api/profit-schedules/create
- GET /api/profit-schedules/list
- GET /api/profit-schedules/{id}
- POST /api/profit-postings/run
- GET /api/profit-postings/list
- GET /api/profit-postings/{id}
- GET /api/profit/dashboard-summary

Business Flow / Workflow

- 128.Profit ratio configured with effective date
- 129.Eligible accounts identified
- 130.Period snapshot taken
- 131.Profit calculated
- 132.Profit posting record created
- 133.Customer statement reflects posting

Important Validations / Business Rules

- Do not overwrite previous effective ratio blindly
- posting_ref unique
- same posting period should not duplicate for same account unless reversal/correction flow exists
- Only eligible accounts should be posted

View Page Requirement

Profit ratio detail, account type, effective window, schedule list, posting history, failed cases.

Action Column Buttons

- View
- Edit ratio
- Create schedule
- Run posting
- View posting detail

Implementation Order for This Module

- 134.Create profit tables
- 135.Profit ratio CRUD
- 136.Profit schedule setup
- 137.Posting run flow
- 138.Posting history
- 139.Profit dashboard

Module 11: Card Management

Category	Details
Sprint/Phase	Sprint 8
Base API path	/api/cards
Goal	Handle card issue, activation, block/unblock, replacement/renewal, PIN events, ATM/CDM linkage and card monitoring.

Dashboard Widgets / Summary Cards

- Total cards
- Active cards
- Blocked cards
- Expiring soon
- Card txn count
- Pending activations
- Card usage alerts today

Tables & Columns

card

- id
- card_ref_no
- customer_id
- account_id
- card_type
- masked_card_no
- issue_date
- expiry_date
- card_status
- block_reason
- status
- created_at

card_event_log

- id
- card_id
- event_type
- event_date
- performed_by
- remarks
- status
- created_at

card_pin_event

- id
- card_id
- event_type
- event_date
- performed_by
- status

- created_at

Forms / Field Source / Save Mapping

Card forms

Field	UI Type	Source	Save Target
Customer	By-search	customer	card.customer_id
Account	By-search	account	card.account_id
Card Type	Dropdown	lookup CARD_TYPE	card.card_type
Masked Card No	System/manual masked	system	card.masked_card_no
Issue/Expiry Date	Date picker	manual/system	card.issue_date/expiry_date
Card Status	Dropdown	lookup CARD_STATUS	card.card_status
Block Reason	Textarea	manual	card.block_reason

Backend Files / Package Guide

- com.sbms.card.controller/ICardController.java
- controller.impl/CardController.java
- service/repository/entity/dto for card, event log, PIN event

Angular Pages & Files

Category	Details
Pages	• card-dashboard • card-list • card-create • card-edit • card-view • card-activation • card-block-unblock • pin-event-list • atm-cdm-transaction-list
Core Angular files	• models/card.model.ts • services/card.service.ts • card-routing.module.ts • card.module.ts

Suggested APIs

- POST /api/cards/create
- GET /api/cards/list
- GET /api/cards/{id}
- PUT /api/cards/{id}
- DELETE /api/cards/{id}
- PUT /api/cards/{id}/restore
- POST /api/cards/{id}/activate
- POST /api/cards/{id}/block
- POST /api/cards/{id}/unblock
- GET /api/cards/{id}/pin-events
- GET /api/cards/atm-cdm-transactions
- GET /api/cards/dashboard-summary

Business Flow / Workflow

140.Customer/account eligibility checked

141.Card request created

- 142.Card record issued
- 143.Card printed/assigned
- 144.PIN event recorded
- 145.Customer activates card
- 146.Usage monitored
- 147.Statement and alerts available

Important Validations / Business Rules

- masked_card_no unique
- only eligible account/customer can get card
- card status controls transaction ability
- non-halal MCC/category restriction should be enforceable where needed
- expiry date required

View Page Requirement

Card profile, customer/account, events, PIN logs, status changes, usage alerts.

Action Column Buttons

- View
- Edit
- Activate
- Block
- Unblock
- Replace
- Renew
- View PIN events

Implementation Order for This Module

- 148.Create card tables
- 149.Card issue flow
- 150.Status/block flow
- 151.PIN event logs
- 152.Card dashboard

Module 12: Customer Statement / Branch Statement

Category	Details
Sprint/Phase	Sprint 9
Base API path	/api/customer-statements, /api/branch-statements
Goal	Provide printable/downloadable account and branch operational summaries with export history.

Dashboard Widgets / Summary Cards

- Statements generated today
- Customer statement requests
- Branch statement requests
- Export/download counts
- Most requested statement types

Tables & Columns

customer_statement_request

- id
- request_no
- customer_id
- account_id
- date_from
- date_to
- request_status
- generated_file_id
- requested_by
- requested_at
- generated_at
- status

branch_statement_request

- id
- request_no
- branch_id
- date_from
- date_to
- request_status
- generated_file_id
- requested_by
- requested_at
- generated_at
- status

file_reference

- id
- file_name
- original_file_name
- file_path
- file_type
- file_size

- module_name
- reference_table
- reference_id
- status
- created_at

Forms / Field Source / Save Mapping

Statement request forms

Field	UI Type	Source	Save Target
Request No	Auto-generated	system	request_no
Customer	By-search	customer	customer_statement_request.customer_id
Account	By-search	account	customer_statement_request.account_id
Branch	Dropdown/search	branch	branch_statement_request.branch_id
Date From/To	Date picker	manual	date_from/date_to
Generated File	System	file_reference	generated_file_id

Backend Files / Package Guide

- com.sbms.statement.controller/ICustomerStatementController.java
- IBranchStatementController.java
- service/repository/entity/dto for statement requests and file references

Angular Pages & Files

Category	Details
Pages	• statement-dashboard • customer-statement-request • customer-statement-list • customer-statement-view • branch-statement-request • branch-statement-list • branch-statement-view • export-center
Core Angular files	• models/statement.model.ts • services/statement.service.ts • statement-routing.module.ts • statement.module.ts

Suggested APIs

- POST /api/customer-statements/request
- GET /api/customer-statements/list
- GET /api/customer-statements/{id}
- GET /api/customer-statements/{id}/download
- POST /api/branch-statements/request
- GET /api/branch-statements/list
- GET /api/branch-statements/{id}
- GET /api/branch-statements/{id}/download
- GET /api/statements/dashboard-summary

Business Flow / Workflow

- 153. User selects account/branch + period
- 154. System fetches transactions/profit/vault/ATM/CDM totals
- 155. Statement lines/summary generated
- 156. PDF/print/download available
- 157. Export action logged

Important Validations / Business Rules

- Date range required
- Account/branch ownership restrictions apply
- Export should be logged for sensitive statements
- Generated file reference required before download

View Page Requirement

Statement request detail, filters, generated file, download/print info, audit metadata.

Action Column Buttons

- View
- Download
- Print
- Export

Implementation Order for This Module

- 158. Create statement tables
- 159. Statement request pages
- 160. Generate export records
- 161. Download/view files
- 162. Statement dashboard

Module 13: Deposit Scheme

Category	Details
Sprint/Phase	Sprint 10
Base API path	/api/deposit-schemes
Goal	Handle scheme definition, enrollment, deposit schedule, profit distribution, maturity and early withdrawal.

Dashboard Widgets / Summary Cards

- Total schemes
- Active enrollments
- Due installments
- Early withdrawal requests
- Matured schemes

Tables & Columns

deposit_scheme

- id
- scheme_code
- scheme_name
- scheme_type
- tenure_months
- minimum_installment
- profit_ratio
- profit_frequency
- status
- created_at

deposit_scheme_enrollment

- id
- enrollment_no
- scheme_id
- customer_id
- linked_account_id
- start_date
- installment_amount
- maturity_date
- enrollment_status
- created_at

deposit_scheme_schedule

- id
- enrollment_id
- installment_no
- due_date
- due_amount
- paid_amount
- paid_date
- schedule_status
- created_at

Forms / Field Source / Save Mapping

Deposit scheme/enrollment forms

Field	UI Type	Source	Save Target
Scheme Code	Auto/text	system/manual	deposit_scheme.scheme_code
Scheme Name	Text box	manual	deposit_scheme.scheme_name
Scheme Type	Dropdown	lookup/static	deposit_scheme.scheme_type
Tenure Months	Number	manual	deposit_scheme.tenure_months
Minimum Installment	Currency	manual	deposit_scheme.minimum_installment
Profit Ratio	Decimal	manual	deposit_scheme.profit_ratio
Profit Frequency	Dropdown	lookup PROFIT_FREQUENCY	deposit_scheme.profit_frequency
Customer	By-search	customer	deposit_scheme_enrollment.customer_id
Linked Account	By-search	account	deposit_scheme_enrollment.linked_account_id
Installment Amount	Currency	manual	deposit_scheme_enrollment.installment_amount

Backend Files / Package Guide

- com.sbms.depositscheme.controller/IDepositSchemeController.java
- controller.impl/DepositSchemeController.java
- service/repository/entity/dto for scheme, enrollment, schedule

Angular Pages & Files

Category	Details
Pages	• scheme-dashboard • scheme-list • scheme-create • scheme-edit • scheme-view • enrollment-list • enrollment-create • enrollment-view • deposit-schedule-view • profit-distribution-view
Core Angular files	• models/deposit-scheme.model.ts • services/deposit-scheme.service.ts • deposit-scheme-routing.module.ts • deposit-scheme.module.ts

Suggested APIs

- POST /api/deposit-schemes/create
- GET /api/deposit-schemes/list
- GET /api/deposit-schemes/{id}
- PUT /api/deposit-schemes/{id}
- DELETE /api/deposit-schemes/{id}
- PUT /api/deposit-schemes/{id}/restore

- POST /api/deposit-schemes/enrollment/create
- GET /api/deposit-schemes/enrollment/list
- GET /api/deposit-schemes/enrollment/{id}
- GET /api/deposit-schemes/enrollment/{id}/schedule
- GET /api/deposit-schemes/enrollment/{id}/profit
- GET /api/deposit-schemes/dashboard-summary

Business Flow / Workflow

- 163.Admin creates scheme master
- 164.Customer enrolls
- 165.Minimum deposit verified
- 166.Schedule created
- 167.Profit distribution runs periodically
- 168.Statement generated
- 169.Maturity/early withdrawal handled

Important Validations / Business Rules

- scheme parameters drive schedule/maturity rules
- minimum installment required
- early withdrawal rule controlled centrally
- scheme contract/version should be traceable

View Page Requirement

Scheme detail, active enrollments, schedule, profit distribution, maturity/early withdrawal status.

Action Column Buttons

- View
- Edit
- Archive/Delete
- Restore
- Enroll
- View schedule
- View profit

Implementation Order for This Module

- 170.Scheme CRUD
- 171.Enrollment create/list/detail
- 172.Schedule page
- 173.Maturity process
- 174.Early withdrawal
- 175.Profit distribution run
- 176.Scheme dashboard

Module 14: Financing

Category	Details
Sprint/Phase	Sprint 11
Base API path	/api/financing-products, /api/financing-applications
Goal	Manage Islamic financing products, applications, asset verification, Shariah review, approval, disbursement, schedule and repayment.

Dashboard Widgets / Summary Cards

- Pending applications
- Approved applications
- Disbursed amount
- Overdue installments
- Financing by product
- Charity late-fee amount

Tables & Columns

financing_product

- id
- product_code
- product_name
- financing_type
- minimum_amount
- maximum_amount
- tenure_months
- profit_rule
- status
- created_at

financing_application

- id
- application_no
- customer_id
- product_id
- branch_id
- requested_amount
- asset_description
- purpose
- application_status
- submitted_at
- approved_by
- approved_at
- remarks
- status
- created_at

financing_asset_verification

- id
- application_id

- asset_value
- verification_note
- verified_by
- verified_at
- status
- created_at

financing_disbursement

- id
- application_id
- disbursement_no
- disbursement_date
- disbursed_amount
- credited_account_id
- disbursed_by
- status
- created_at

financing_schedule

- id
- application_id
- installment_no
- due_date
- principal_amount
- profit_amount
- charity_amount
- paid_amount
- paid_date
- schedule_status
- created_at

Forms / Field Source / Save Mapping

Financing product/application forms

Field	UI Type	Source	Save Target
Product Code	Auto/text	system/manual	financing_product.product_code
Product Name	Text box	manual	financing_product.product_name
Financing Type	Dropdown	lookup FINANCING_TYPE	financing_product.financing_type
Minimum/Maximum Amount	Currency	manual	financing_product.minimum_amount/ maximum_amount
Tenure Months	Number	manual	financing_product.tenure_months
Profit Rule	Textarea/dropdown	manual	financing_product.profit_rule
Application No	Auto	system	financing_application.application_no
Customer	By-search	customer	financing_application.customer_id

Branch	Dropdown/search	branch	financing_application.branch_id
Requested Amount	Currency	manual	financing_application.requested_amount
Asset Description	Textarea	manual	financing_application.asset_description
Purpose	Textarea	manual	financing_application.purpose

Backend Files / Package Guide

- com.sbms.financing.controller/IFinancingProductController.java
- IFinancingApplicationController.java
- service/repository/entity/dto for product, application, asset verification, disbursement, schedule

Angular Pages & Files

Category	Details
Pages	• financing-dashboard • financing-product-list • financing-product-create • financing-product-view • financing-application-list • financing-application-create • financing-application-edit • financing-application-view • financing-review • financing-disbursement • installment-schedule-view • repayment-collection
Core Angular files	• models/financing.model.ts • services/financing.service.ts • financing-routing.module.ts • financing.module.ts

Suggested APIs

- POST /api/financing-products/create
- GET /api/financing-products/list
- GET /api/financing-products/{id}
- PUT /api/financing-products/{id}
- DELETE /api/financing-products/{id}
- POST /api/financing-applications/create
- GET /api/financing-applications/list
- GET /api/financing-applications/{id}
- PUT /api/financing-applications/{id}
- POST /api/financing-applications/{id}/submit
- POST /api/financing-applications/{id}/verify
- POST /api/financing-applications/{id}/approve
- POST /api/financing-applications/{id}/reject
- POST /api/financing-applications/{id}/return
- POST /api/financing-applications/{id}/disburse
- GET /api/financing-applications/{id}/schedule
- POST /api/financing-applications/{id}/collect-payment
- GET /api/financing/dashboard-summary

Business Flow / Workflow

- 177.Product exists
- 178.Customer submits financing request
- 179.Officer collects documents
- 180.Purpose and asset validated
- 181.Asset verification done
- 182.Cost + profit determined
- 183.Contract generated
- 184.Shariah board review
- 185.Branch manager/operations approval
- 186.Disbursement made
- 187.Installment schedule created
- 188.Repayments collected

Status flow: DRAFT -> SUBMITTED -> DOC_CHECK -> ASSET_VERIFIED -> SHARIAH_REVIEW -> APPROVED -> DISBURSED -> ACTIVE -> CLOSED

Important Validations / Business Rules

- No interest logic
- product-specific formula applies
- asset-required products need asset data
- disbursement should not happen before approval chain complete
- late fee goes to charity only
- requested amount within product min/max

View Page Requirement

Application detail, product, customer, asset verification, approvals, disbursement, schedule, repayments.

Action Column Buttons

- View
- Edit
- Submit
- Verify
- Approve
- Reject
- Return
- Disburse
- Collect payment

Implementation Order for This Module

- 189.Product CRUD
- 190.Application create/list/detail/update
- 191.Submit
- 192.Asset verification save
- 193.Shariah submit
- 194.Approve/reject/return
- 195.Disburse
- 196.Schedule list
- 197.Repayment collection

Module 15: Contract Management

Category	Details
Sprint/Phase	Sprint 12
Base API path	/api/contracts
Goal	Manage contract templates, generated contracts, e-sign flow, versioning, locking and archive.

Dashboard Widgets / Summary Cards

- Total contracts
- Pending signatures
- Active/locked contracts
- Contract versions
- Draft contracts

Tables & Columns

contract_template

- id
- template_code
- template_name
- contract_type
- version_no
- template_body
- status
- created_at

contract

- id
- contract_no
- contract_type
- template_id
- customer_id
- reference_module
- reference_id
- contract_text
- signed_by_customer
- signed_by_shariah
- signed_date
- contract_status
- status
- created_at

Forms / Field Source / Save Mapping

Contract forms

Field	UI Type	Source	Save Target
Template Code	Auto/text	system/manual	contract_template.template_code
Template Name	Text box	manual	contract_template.template_name

Contract Type	Dropdown	lookup/static	contract_template.contract_type
Version No	Number/text	manual/system	contract_template.version_no
Template Body	Rich textarea	manual	contract_template.template_body
Contract No	Auto	system	contract.contract_no
Customer	By-search	customer	contract.customer_id
Reference Module/ID	Derived	source module	contract.reference_module/reference_id
Contract Text	Generated textarea	system/editable before lock	contract.contract_text

Backend Files / Package Guide

- com.sbms.contract.controller/IContractController.java
- service/repository/entity/dto for template, contract, versions

Angular Pages & Files

Category	Details
Pages	• contract-dashboard • contract-template-list • contract-template-create • contract-template-view • contract-list • contract-generate • contract-view • contract-sign • contract-version-history
Core Angular files	• models/contract.model.ts • services/contract.service.ts • contract-routing.module.ts • contract.module.ts

Suggested APIs

- POST /api/contracts/templates/create
- GET /api/contracts/templates/list
- GET /api/contracts/templates/{id}
- PUT /api/contracts/templates/{id}
- POST /api/contracts/generate
- GET /api/contracts/list
- GET /api/contracts/{id}
- POST /api/contracts/{id}/customer-sign
- POST /api/contracts/{id}/shariah-sign
- GET /api/contracts/{id}/versions
- GET /api/contracts/dashboard-summary

Business Flow / Workflow

- 199.Contract template exists
- 200.System auto-fills customer/asset/deal data
- 201.Shariah clauses inserted
- 202.Officer reviews
- 203.Customer e-sign
- 204.Shariah board e-sign if applicable

205.Contract locked

206.Repository stores final and old versions

Important Validations / Business Rules

- Locked contract should not be edited directly
- New changes create new version if needed
- Signer identity/time must be stored
- Template code unique

View Page Requirement

Contract text, sign status, signer identity/time, template version, source module, version history.

Action Column Buttons

- View
- Edit template
- Generate
- Sign
- View versions
- Download

Implementation Order for This Module

207.Contract template CRUD

208.Contract generate

209.Contract list/detail

210.Customer sign

211.Shariah sign

212.Version list

213.Dashboard

Module 16: Shariah Review

Category	Details
Sprint/Phase	Sprint 13
Base API path	/api/shariah-reviews
Goal	Support Shariah compliance checklist review, approval/reject/correction, certification and audit trail.

Dashboard Widgets / Summary Cards

- Pending cases
- Approved cases
- Rejected cases
- Correction requests
- Upcoming meetings/reviews if implemented

Tables & Columns

shariah_review_case

- id
- case_no
- reference_module
- reference_id
- submitted_by
- submitted_at
- case_status
- remarks
- status
- created_at

shariah_checklist_item

- id
- item_code
- item_name
- description
- status
- created_at

shariah_review_decision

- id
- case_id
- decision
- decision_by
- decision_at
- remarks
- status
- created_at

Forms / Field Source / Save Mapping

Shariah review forms

Field	UI Type	Source	Save Target
-------	---------	--------	-------------

Case No	Auto	system	shariah_review_case.case_no
Reference Module/ID	Derived	source module	reference_module/reference_id
Submitted By/At	System	logged-in user/time	submitted_by/submitted_at
Checklist Items	Checklist	shariah_checklist_item	decision/checklist records
Decision	Dropdown	APPROVE/REJECT/RETURN	shariah_review_decision.decision
Remarks	Textarea	manual	remarks

Backend Files / Package Guide

- com.sbms.shariah.controller/IShariahReviewController.java
- service/repository/entity/dto for case, checklist, decision

Angular Pages & Files

Category	Details
Pages	• shariah-dashboard • case-list • case-view • case-review • correction-queue • fatwa-certificate-list • annual-report-view
Core Angular files	• models/shariah.model.ts • services/shariah.service.ts • shariah-routing.module.ts • shariah.module.ts

Suggested APIs

- POST /api/shariah-reviews/create
- GET /api/shariah-reviews/list
- GET /api/shariah-reviews/{id}
- POST /api/shariah-reviews/{id}/approve
- POST /api/shariah-reviews/{id}/reject
- POST /api/shariah-reviews/{id}/return
- GET /api/shariah-reviews/{id}/history
- GET /api/shariah-reviews/dashboard-summary

Business Flow / Workflow

214. Officer submits case
215. Checklist prepared
216. Board reviews purpose/asset/contract/docs
217. Board decides approve/reject/correction
218. Decision stored
219. Annual report uses data

Important Validations / Business Rules

- Board decision mandatory before certain financing/contracts proceed
- Correction remarks mandatory when returned
- All decisions need traceability
- Checklist items should be active

View Page Requirement

Case source, checklist, review notes, decisions, reviewer identity, history.

Action Column Buttons

- View
- Approve
- Reject
- Return
- History

Implementation Order for This Module

220.Review case create/list/detail

221.Checklist load/save

222.Approve

223.Reject

224.Correction/return

225.History

226.Dashboard

Module 17: Zakat & Charity

Category	Details
Sprint/Phase	Sprint 14
Base API path	/api/zakat
Goal	Calculate zakat, handle deduction, charity fund ledger, beneficiaries and payouts.

Dashboard Widgets / Summary Cards

- Zakat due accounts
- Total zakat calculated
- Charity fund balance
- Beneficiary payout total
- Upcoming zakat reminders

Tables & Columns

zakat_profile

- id
- customer_id
- zakat_year
- nisab_amount
- eligible_asset_amount
- zakat_amount
- calculation_status
- created_at

charity_fund

- id
- fund_date
- source_type
- reference_id
- credit_amount
- debit_amount
- balance_after
- remarks
- created_at

charity_beneficiary

- id
- beneficiary_code
- beneficiary_name
- mobile
- address
- status
- created_at

charity_payout

- id
- beneficiary_id
- payout_date

- amount
- approved_by
- remarks
- status
- created_at

Forms / Field Source / Save Mapping

Zakat/charity forms

Field	UI Type	Source	Save Target
Customer	By-search	customer	zakat_profile.customer_id
Zakat Year	Number	manual	zakat_profile.zakat_year
Nisab Amount	Currency	manual/system	zakat_profile.nisab_amount
Eligible Asset Amount	Currency	system/manual	zakat_profile.eligible_asset_amount
Beneficiary Code/Name	Text	manual	charity_beneficiary
Payout Date/Amount	Date/currency	manual	charity_payout.payout_date/ amount
Remarks	Textarea	manual	remarks

Backend Files / Package Guide

- com.sbms.zakat.controller/IZakatController.java
- service/repository/entity/dto for profile, charity fund, beneficiary, payout

Angular Pages & Files

Category	Details
Pages	• zakat-dashboard • zakat-profile-list • zakat-profile-view • zakat-calc-run • charity-fund-view • beneficiary-list • beneficiary-create • payout-list • payout-create
Core Angular files	• models/zakat.model.ts • services/zakat.service.ts • zakat-routing.module.ts • zakat.module.ts

Suggested APIs

- POST /api/zakat/profile/create
- GET /api/zakat/profile/list
- GET /api/zakat/profile/{id}
- POST /api/zakat/calculate
- GET /api/zakat/charity-fund
- GET /api/zakat/beneficiaries/list
- POST /api/zakat/beneficiaries/create
- GET /api/zakat/payouts/list
- POST /api/zakat/payouts/create
- GET /api/zakat/dashboard-summary

Business Flow / Workflow

- 227.Customer opts in/profile maintained
- 228.Eligible zakatable assets/balance determined
- 229.Yearly calculation executed
- 230.Deduction processed if applicable
- 231.Charity fund ledger updated
- 232.Beneficiary payout managed
- 233.Reports generated

Important Validations / Business Rules

- Asset-based calculation only
- Deduction traceable by year/period
- Late fees/penalties route to charity fund only
- Beneficiary must be active for payout

View Page Requirement

Zakat profile, calculation amount, charity fund movements, payout details.

Action Column Buttons

- View
- Calculate
- Add beneficiary
- Payout
- Export report

Implementation Order for This Module

- 234.Profile create/list/update
- 235.Calculate
- 236.Deduction run
- 237.Charity fund ledger
- 238.Beneficiary CRUD
- 239.Payout create/list
- 240.Report export
- 241.Dashboard

Module 18: Notification & Alerts

Category	Details
Sprint/Phase	Sprint 15
Base API path	/api/notifications
Goal	Convert system events into SMS/email/push messages and store delivery logs/retry queue.

Dashboard Widgets / Summary Cards

- Sent today
- Failed today
- Retry queue
- Channel wise summary
- Messages sent today
- Failed deliveries

Tables & Columns

notification_template

- id
- template_code
- template_name
- channel_type
- subject_text
- body_text
- status
- created_at

notification_event

- id
- event_code
- event_name
- reference_module
- status
- created_at

notification_log

- id
- event_id
- template_id
- recipient_to
- channel_type
- delivery_status
- provider_response
- retry_count
- sent_at
- status
- created_at

Forms / Field Source / Save Mapping

Notification forms

Field	UI Type	Source	Save Target
Template Code/Name	Text	manual	notification_template
Channel Type	Dropdown	EMAIL/SMS/PUSH	notification_template.channel_type
Subject Text	Text	manual	notification_template.subject_text
Body Text	Textarea	manual	notification_template.body_text
Event Code/Name	Text	manual	notification_event
Recipient	Text/system	derived	notification_log.recipient_to

Backend Files / Package Guide

- com.sbms.notification.controller/INotificationController.java
- service/repository/entity/dto for template, event, log

Angular Pages & Files

Category	Details
Pages	• notification-dashboard • template-list • template-create • template-view • event-rule-list • event-rule-create • delivery-log-list • delivery-log-view • retry-queue
Core Angular files	• models/notification.model.ts • services/notification.service.ts • notification-routing.module.ts • notification.module.ts

Suggested APIs

- POST /api/notifications/templates/create
- GET /api/notifications/templates/list
- GET /api/notifications/templates/{id}
- PUT /api/notifications/templates/{id}
- GET /api/notifications/event-rules/list
- POST /api/notifications/event-rules/create
- GET /api/notifications/logs/list
- GET /api/notifications/logs/{id}
- POST /api/notifications/logs/{id}/retry
- GET /api/notifications/dashboard-summary

Business Flow / Workflow

242. Business event happens
243. Notification event created
244. Template resolved
245. Rules determine channel(s)
246. Provider called

247.Delivery result logged

248.Retry queued if failed

Important Validations / Business Rules

- Do not mix messaging code directly in business modules; call notification service
- Template centrally managed
- Failed sends remain traceable
- Retry count controlled

View Page Requirement

Template/event/log detail, provider response, retry count and delivery status.

Action Column Buttons

- View
- Edit template
- Retry
- Archive

Implementation Order for This Module

249.Template CRUD

250.Event mapping

251.Send test

252.Logs list/detail

253.Retry

254.Dashboard

Module 19: Integration Management

Category	Details
Sprint/Phase	Sprint 15
Base API path	/api/integrations
Goal	Store provider configuration for SMS/email/payment/mobile banking and log execution/retry status.

Dashboard Widgets / Summary Cards

- Active providers
- Failed integrations
- Retry pending
- Last successful sync
- Success rate

Tables & Columns

integration_provider

- id
- provider_code
- provider_name
- provider_type
- base_url
- auth_type
- api_key
- username
- password_enc
- timeout_sec
- status
- created_at

integration_execution_log

- id
- provider_id
- reference_module
- reference_id
- request_payload
- response_payload
- http_status
- execution_status
- executed_at
- retry_count
- status
- created_at

Forms / Field Source / Save Mapping

Integration provider/log forms

Field	UI Type	Source	Save Target
Provider Code/Name	Text	manual	integration_provider

Provider Type	Dropdown	EMAIL/SMS/PAYMENT/etc	integration_provider.provider_type
Base URL	Text	manual	integration_provider.base_url
Auth Type	Dropdown	static	integration_provider.auth_type
API Key/Username/Password	Text/password	manual secured	integration_provider
Timeout Sec	Number	manual	integration_provider.timeout_sec

Backend Files / Package Guide

- com.sbms.integration.controller/IIntegrationController.java
- service/repository/entity/dto for provider and execution log

Angular Pages & Files

Category	Details
Pages	• integration-dashboard • provider-list • provider-create • provider-edit • provider-view • integration-log-list • integration-log-view • provider-test
Core Angular files	• models/integration.model.ts • services/integration.service.ts • integration-routing.module.ts • integration.module.ts

Suggested APIs

- POST /api/integrations/providers/create
- GET /api/integrations/providers/list
- GET /api/integrations/providers/{id}
- PUT /api/integrations/providers/{id}
- DELETE /api/integrations/providers/{id}
- PUT /api/integrations/providers/{id}/restore
- POST /api/integrations/providers/{id}/test
- GET /api/integrations/logs/list
- GET /api/integrations/logs/{id}
- GET /api/integrations/dashboard-summary

Business Flow / Workflow

- 255.Provider configured
- 256.Module triggers integration request
- 257.Gateway/provider selected
- 258.Request sent
- 259.Response stored
- 260.Retry/monitoring done if needed

Important Validations / Business Rules

- Provider can be active/inactive

- Credentials/config must be masked or secured
- Delivery/execution logs mandatory
- Timeout required

View Page Requirement

Provider config with masked credentials, test status, logs and retry history.

Action Column Buttons

- View
- Edit
- Archive/Delete
- Restore
- Test provider
- View logs

Implementation Order for This Module

261.Provider CRUD

262.Provider test connection

263.Execution logs

264.Retry execution

265.Dashboard

Module 20: Reporting & Regulatory

Category	Details
Sprint/Phase	Sprint 16
Base API path	/api/reports
Goal	Provide operational, regulatory, financing, profit, PAR, Shariah audit and branch reports with export history.

Dashboard Widgets / Summary Cards

- Reports generated today
- Regulatory pending
- Branch performance summary
- Financing summary
- Profit summary
- Most used reports

Tables & Columns

report_definition

- id
- report_code
- report_name
- report_type
- query_key
- export_types
- status
- created_at

report_request_log

- id
- report_id
- requested_by
- date_from
- date_to
- filter_json
- generated_file_id
- request_status
- requested_at
- generated_at
- status

Forms / Field Source / Save Mapping

Report forms

Field	UI Type	Source	Save Target
Report Code/Name	Text	manual	report_definition
Report Type	Dropdown	static/lookup	report_definition.report_type
Query Key	Text	system/manual	report_definition.query_key
Export Types	Checkbox/multi	PDF/CSV/Excel	report_definition.export_types

Date From/To	Date picker	manual	report_request_log.date_from/date_to
Filter JSON	System/manual	filters	report_request_log.filter_json

Backend Files / Package Guide

- com.sbms.report.controller/IReportController.java
- service/repository/entity/dto for definition and request log

Angular Pages & Files

Category	Details
Pages	• report-dashboard • operational-report • profit-distribution-report • financing-portfolio-report • par-report • shariah-audit-report • branch-report • export-history
Core Angular files	• models/report.model.ts • services/report.service.ts • report-routing.module.ts • report.module.ts

Suggested APIs

- GET /api/reports/operational
- GET /api/reports/profit-distribution
- GET /api/reports/financing-portfolio
- GET /api/reports/par
- GET /api/reports/shariah-audit
- GET /api/reports/branch
- GET /api/reports/export-history
- GET /api/reports/dashboard-summary

Business Flow / Workflow

266. User selects report type
267. User selects filters/date range
268. Data fetched and aggregated
269. Report shown in dashboard/table
270. Export PDF/CSV/print
271. Export log stored

Important Validations / Business Rules

- Report queries must use clear filters
- Sensitive reports need role restrictions
- Export action should be auditable
- Date range validation required

View Page Requirement

Report result, filters, generated file, run status, export history.

Action Column Buttons

- View
- Filter
- Export PDF
- Export CSV
- Print

Implementation Order for This Module

272. Definition CRUD

273. Operational run

274. Regulatory run

275. Export PDF/CSV

276. History

277. Dashboard

Module 21: Security / Audit / Investigation

Category	Details
Sprint/Phase	Sprint 17
Base API path	/api/security-events, /api/audit-logs, /api/investigations
Goal	Track failed logins, audit events, suspicious activities, AML/sanction hooks and investigation lifecycle.

Dashboard Widgets / Summary Cards

- Failed logins today
- Locked users
- Suspicious txn count
- Audit events today
- Open investigation cases
- AML flags today

Tables & Columns

security_event_log

- id
- event_code
- event_name
- user_id
- ip_address
- device_info
- reference_module
- reference_id
- event_time
- severity_level
- remarks
- status

audit_log

- id
- module_name
- reference_id
- action_name
- old_value_json
- new_value_json
- performed_by
- performed_at
- status

investigation_case

- id
- case_no
- case_type
- reference_module
- reference_id
- opened_by
- opened_at

- assigned_to
- case_status
- remarks
- status
- created_at

Forms / Field Source / Save Mapping

Security/investigation forms

Field	UI Type	Source	Save Target
Event Code/Name	System	security event	security_event_log
Module/Reference	System	source action	reference_module/ reference_id
Severity Level	Dropdown	static	security_event_log.severity_level
Case No	Auto	system	investigation_case.case_no
Case Type	Dropdown	static/lookup	investigation_case.case_type
Assigned To	By-search	app_user	investigation_case.assigned_to
Remarks	Textarea	manual	remarks

Backend Files / Package Guide

- com.sbms.security.controller/ISecurityController.java
- IAuditLogController.java
- IInvestigationController.java
- service/repository/entity/dto for event, audit, investigation

Angular Pages & Files

Category	Details
Pages	• security-dashboard • security-event-list • security-event-view • suspicious-activity-list • suspicious-activity-view • investigation-case-list • investigation-case-view • investigation-case-action
Core Angular files	• models/security.model.ts • services/security.service.ts • security-routing.module.ts • security.module.ts

Suggested APIs

- GET /api/security/events/list
- GET /api/security/events/{id}
- GET /api/security/suspicious-activities/list
- GET /api/security/suspicious-activities/{id}
- GET /api/security/investigation-cases/list
- GET /api/security/investigation-cases/{id}
- POST /api/security/investigation-cases/{id}/assign
- POST /api/security/investigation-cases/{id}/close
- GET /api/security/dashboard-summary

Business Flow / Workflow

- 278. User logs in or transaction occurs
- 279. Security/compliance rules scan event
- 280. Suspicious activity may be raised
- 281. Investigation case created
- 282. Admin/compliance officer reviews
- 283. Decision/closure stored

Important Validations / Business Rules

- Suspicious cases must have lifecycle and comments
- High-risk events must not disappear from view
- AML/sanction hooks should be modular
- Sensitive actions must be logged

View Page Requirement

Security event detail, audit diff, investigation lifecycle, assigned user and closure remarks.

Action Column Buttons

- View
- Assign
- Close
- Export

Implementation Order for This Module

- 284. Security event list/detail
- 285. Audit log list/detail
- 286. Investigation create/list/detail/update/assign/close
- 287. Dashboard

Module 22: Workflow Support

Category	Details
Sprint/Phase	Sprint 18
Base API path	/api/workflows
Goal	Provide generic workflow history, comments, pending approval queues and submission tracking.

Dashboard Widgets / Summary Cards

- Pending approvals
- My pending tasks
- Recent completed tasks
- Workflow bottlenecks

Tables & Columns

workflow_history

- id
- module_name
- reference_id
- action_name
- from_status
- to_status
- action_by
- action_at
- remarks
- status

workflow_comment

- id
- module_name
- reference_id
- comment_text
- comment_by
- comment_at
- status

Forms / Field Source / Save Mapping

Workflow/comment forms

Field	UI Type	Source	Save Target
Module Name	System	source module	workflow_history.module_name
Reference ID	System	source record	workflow_history.reference_id
Action Name	Dropdown/system	workflow action	workflow_history.action_name
From/To Status	System	status transition	workflow_history.from_status/to_status
Remarks/Comment	Textarea	manual	workflow_history.remarks/workflow_comment.comment_text

Backend Files / Package Guide

- `com.sbms.workflow.controller/IWorkflowController.java`
- `service/repository/entity/dto` for workflow history and comments

Angular Pages & Files

Category	Details
Pages	• <code>workflow-dashboard</code> • <code>workflow-history-list</code> • <code>workflow-history-view</code> • <code>pending-approval-queue</code> • <code>my-submissions</code>
Core Angular files	• <code>models/workflow.model.ts</code> • <code>services/workflow.service.ts</code> • <code>workflow-routing.module.ts</code> • <code>workflow.module.ts</code>

Suggested APIs

- `GET /api/workflows/history/list`
- `GET /api/workflows/history/{id}`
- `GET /api/workflows/pending`
- `GET /api/workflows/my-submissions`
- `GET /api/workflows/dashboard-summary`

Business Flow / Workflow

- 288.Source module creates/updates status
- 289.Workflow history row inserted
- 290.Pending queues query actionable items
- 291.Users review/approve/reject/return
- 292.History/timeline remains traceable

Important Validations / Business Rules

- Every workflow transition must store actor and timestamp
- Invalid direct jumps should be blocked
- Workflow comments/history normally should not be deleted
- Source record must exist

View Page Requirement

Workflow action record, source module, old/new status, action actor, remarks, related comments.

Action Column Buttons

- View
- Open source record
- Export

Implementation Order for This Module

- 293.Workflow history list/view
- 294.Pending approvals
- 295.My submissions
- 296.Comments/timeline
- 297.Dashboard

Add-on Module: Verification / OTP / Email / SMS

Category	Details
Sprint/Phase	Phase 0 + Sprint 1-2 + Sprint 4 + Sprint 15 + Sprint 17
Base API path	/api/verification plus auth/customer/notification/integration hooks
Goal	Support user/customer email verification, mobile verification, login OTP, password reset OTP and critical action verification.

Dashboard Widgets / Summary Cards

- Auth: pending password resets, OTP sent/failed today, locked users due to OTP failure
- Customer: unverified mobile/email count
- Notification: email/SMS sent/failed, queue pending, retry pending
- Integration: active providers, sandbox providers, provider failure rate
- Security: OTP failure spike, suspicious login attempts

Tables & Columns

verification_channel

- id
- channel_code
- channel_name
- provider_type
- is_otp_channel
- is_active
- status
- created_at
- updated_at

verification_template

- id
- template_code
- template_name
- channel_id
- module_name
- subject_text
- body_text
- otp_length
- expiry_minutes
- max_retry_count
- is_active
- status
- created_at
- updated_at

contact_verification_status

- id
- reference_module
- reference_id
- contact_type

- contact_value
- is_primary
- is_verified
- verified_at
- verified_by
- verification_method
- last_verification_request_id
- status
- created_at
- updated_at

otp_verification_request

- id
- request_no
- reference_module
- reference_id
- purpose_code
- channel_type
- template_id
- provider_id
- recipient_name
- contact_type
- contact_value
- country_code
- otp_code_hash
- otp_length
- sent_message_id
- sent_at
- expires_at
- verified_at
- attempt_count
- max_attempt_count
- resend_count
- max_resend_count
- request_status
- provider_response
- failure_reason
- ip_address
- device_info
- status
- created_at
- updated_at

verification_dispatch_queue

- id
- reference_module
- reference_id
- request_id
- template_id
- provider_id

- channel_type
- recipient_to
- subject_text
- body_text
- payload_json
- queue_status
- retry_count
- next_retry_at
- last_error
- locked_by
- locked_at
- processed_at
- status
- created_at
- updated_at

password_reset_request

- id
- request_no
- user_id
- reference_value
- channel_type
- provider_id
- token_code_hash
- token_type
- sent_at
- expires_at
- used_at
- attempt_count
- request_status
- provider_response
- status
- created_at
- updated_at

verification_attempt_log

- id
- request_id
- attempt_type
- attempt_value_masked
- attempt_status
- remarks
- ip_address
- device_info
- created_by
- created_at

Forms / Field Source / Save Mapping

Verification-related forms

Field	UI Type	Source	Save Target
User create verification flags	Checkbox	static	verification_dispatch_queue / app_user flags
Customer mobile/email verification buttons	Action button	customer contact	otp_verification_request.contact_value
Login OTP code	6-digit input	manual	verification request validation only
Forgot password identifier/channel	Text/radio	manual	password_reset_request
Admin verification console filters	Date/module/channel/status/provider	logs	verification logs

Backend Files / Package Guide

- com.sbms.verification.controller/VerificationController.java
- service/VerificationService.java
- repository for channel/template/contact/OTP/queue/password reset/attempt log
- entity package for all verification tables
- enums: ChannelType, VerificationPurpose, VerificationStatus, TokenType
- util: OtpGeneratorUtil, OtpHashUtil, MaskingUtil

Angular Pages & Files

Category	Details
Pages	• verification-dashboard • otp-verify • forgot-password • reset-password • verification-log-list • provider-test-console
Core Angular files	• models/otp-request.model.ts • models/verification-log.model.ts • models/verification-template.model.ts • services/verification.service.ts • verification-routing.module.ts • verification.module.ts

Suggested APIs

- POST /api/auth/send-otp
- POST /api/auth/verify-otp
- POST /api/auth/forgot-password
- POST /api/auth/reset-password
- Verification module APIs: sendEmailVerificationOtp, sendMobileVerificationOtp, verifyOtp, resendOtp, expireOtp, sendPasswordResetToken

Business Flow / Workflow

298. User/customer/contact action requires verification
299. OTP request created
300. Dispatch queue row created
301. Provider sends email/SMS or local log is stored
302. User submits OTP

- 303.System validates hash/expiry/attempts
- 304.Verified flags updated
- 305.Attempt log inserted

Important Validations / Business Rules

- email format
- mobile format
- duplicate email/mobile
- resend cooldown seconds
- expiry time not crossed
- attempt count not exceeded
- user not locked
- customer not blocked
- correct module + reference id exists before sending OTP
- one active unexpired OTP at a time per purpose per contact

View Page Requirement

Verification admin console with request, queue, attempt, provider response and contact verification status.

Action Column Buttons

- Send OTP
- Verify OTP
- Resend
- Expire
- Mark failed
- Inspect provider response
- Export logs

Implementation Order for This Module

- 306.Create verification tables
- 307.Seed channels/purposes/templates
- 308.Integrate user create flow
- 309.Integrate customer onboarding flow
- 310.Implement send/verify/resend/expire methods
- 311.Add provider adapter layer
- 312.Update verified flags after success
- 313.Secure abuse points

Support Concept: Calculation Engine

Category	Details
Sprint/Phase	Advanced support for Financing/Profit/Deposit Scheme
Base API path	Internal service / optional /api/calculations
Goal	Provide formula engine for Murabaha, Ijara, Musharaka/Mudaraba P&L, installment planning, restructuring, partial settlement and early payment.

Dashboard Widgets / Summary Cards

- Recent calculations
- Failed calculations
- Product-wise simulation counts

Tables & Columns

No dedicated table listed in handbook; usually implemented as shared/internal service or source-module records.

Forms / Field Source / Save Mapping

Calculation inputs

Field	UI Type	Source	Save Target
Product Type	Dropdown	financing/deposit product	formula selection
Principal/Cost/Asset Value	Currency	manual/source module	calculation input
Profit/Markup/Rental Rule	Decimal/rule	product	calculation input
Tenure/Frequency	Number/dropdown	manual/product	schedule input

Backend Files / Package Guide

- com.sbms.calculation.service/CalculationEngineService.java
- Formula strategy/helper classes
- DTOs for calculation request/response

Angular Pages & Files

Category	Details
Pages	• Optional calculation simulator • Used inside financing/product/schedule pages
Core Angular files	• models/calculation.model.ts • services/calculation.service.ts

Suggested APIs

- Optional POST /api/calculations/simulate
- Internal service calls from financing/profit/deposit modules

Business Flow / Workflow

314. Business module sends parameters
315. Engine selects formula by product
316. Values calculated
317. Schedule/result returned
318. Results stored in source module table

Important Validations / Business Rules

- Do not hardcode interest logic

- Formula must depend on Islamic product type
- Schedule must be reproducible for audit
- Inputs must be positive and business-valid

View Page Requirement

Simulation/result details inside source module.

Action Column Buttons

- Calculate
- Preview schedule
- Apply result

Implementation Order for This Module

319. Define formula rules

320. Create calculation service

321. Return schedule/result

322. Store result in source module table

Appendix A: Standard Frontend Foundation

- Bootstrap + SCSS enabled
- Font Awesome 4 integrated
- SweetAlert2 integrated
- Shared DataTable strategy with search/filter/pagination/export
- Global navigation search
- AuthInterceptor, ErrorInterceptor, optional LoadingInterceptor
- AuthGuard and PermissionGuard
- Core API service, Alert service, Storage/token utility
- Shared page header, page title, status badge, table actions, dashboard card, empty/loading state
- Topbar, Sidebar, Layout shell
- Export/print helper

Appendix B: Standard Soft Delete Decision

Category	Details
Soft delete yes	role, user, lookup value, branch, terminal, customer, address, identity, kyc profile, account type, card, deposit scheme, financing product/application, contract/template, notification template, integration provider, report definition, beneficiary
No hard delete / keep history	transaction journal, cash transaction, fund transfer, transaction reversal, profit posting, audit log, security event log, workflow history, user session, notification log, integration execution log, report request log, charity payout, financing disbursement
Use status instead of delete	account, vault balance, teller limit, profit schedule, statement request, shariah decision, investigation case
Button wording	Role/User/Branch/Template = Archive; Customer/Card/Product = Deactivate; Account = Block/Freeze/Close; Transaction = Reverse

Appendix C: Universal Business Mindset

- Not just CRUD - think workflow.
- Not just data save - think role and approval.
- Not just form - think dashboard/report/statement impact.
- Not just backend - think frontend queue and action buttons.
- Not just success path - think reject/correction/failure path.
- Not just one table - think main + detail + workflow + log tables.
- Not just visible menu - think backend permission check too.
- Not just financial output - think Shariah and audit trace too.